Project Report

"Green Synthesis of Iron oxide Nanoparticles using Neem Leaf Extract"

February to April (2026)

by

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Introduction

Nanoparticle refers to a minute particle having particle sizes from 1 to 100 nanometers. The minute particles have special properties which are significantly different from properties exhibited by the same material in its macro-scale form. Some of the special properties include enhanced optical, electrical, magnetic, and chemical properties. This is due to the large surface area possessed by the nanoparticle and quantum mechanical effects.

Background

Nanoparticles have become important over the last few years in several fields such as medicine, environment, electronics, and energy. In the case of the medicine field, there are uses of nanoparticles in drug delivery and antimicrobial coating. Nanoparticles are also gaining importance in environmental sciences as they are useful in purifying water. Moreover, in recent times, nanoparticles have been increasingly utilized in manufacturing energy devices like solar panels and fuel cells. However, traditional methods of nanoparticle production require the use of toxic agents. Production of nanoparticles uses significant amounts of energy and damages the environment in the process. This has made it necessary to develop a new way of manufacturing nanoparticles. Scientists have therefore developed a novel method of nanoparticle production known as the green synthesis method.Green synthesis has been found to be an environment-friendly and economical technique used in preparing nanoparticles. This technique uses biological sources like extracts from plants, microorganisms such as bacteria and fungi, and biomolecules for reducing and stabilizing nanoparticles. The extracts of plants such as Neem extract have phytochemical compounds like flavonoids, alkaloids, and polyphenols that act as reducing and stabilizing agents. The technique comprises three major stages of reducing metal ions, nanoparticle growth, and stabilization or capping.

Literature Review:

As it was mentioned by some scientific studies associated with this subject, plants prove to be useful when it comes to manufacturing metal oxide nanoparticles. It should be noted that extracts obtained from plants, for instance, consist of many bioactive compounds. As for the neem extract, it contains flavonoids, polyphenols, and enzymes. In addition, these compounds act as reducing and stabilizing agents. Moreover, multi-metal oxide nanoparticles allow combining different abilities of magnetic, catalytic, and antimicrobial nanoparticles. The most popular method for characterization of nanoparticles is UV-Visible spectroscopy because of Surface Plasmon Resonance (SPR).

Objectives of the Study

The main goal of this physics project is to know an environment-friendly approach for preparing multi-metal oxide nanoparticles via Neem leaf extract under the guidelines of Green Chemistry. The use of Neem extract as a reductant will ensure low environmental pollution during the process of preparing nanoparticles.

Specific objectives of the study include:

- Preparing iron oxide nanoparticles utilizing Neem extract as a natural reductant.
- Employing a mixture of different metals such as Iron (Fe), Cobalt (Co), Zinc (Zn), Magnesium (Mg), Copper (Cu), and Nickel (Ni) salts in order to explore any synergistic effects of these substances in nanoparticles produced.
- Applying the necessary conditions for preparation, such as pH regulation and heating (ripening).
- Characterizing the prepared nanoparticles with the help of UV-visible spectroscopy and confirming their existence with a help of optical analysis.
- Performing the analysis of the absorption spectrum and identifying the scientific explanation for the

peaks identified, especially for the Surface Plasmon Resonance (SPR) peak.

In summary, this research will try to determine the basic concepts behind the synthesis of nanoparticles using an environmentally friendly technique, along with understanding their physical nature and possible uses.

Methodology: Experimental Procedure

The process strictly adhered to the green synthesis procedure, which involved Neem leaf extract as the bioreactor. The process comprised of three major stages; extraction, inclusion of precursors and thermal processing.

Preparing Neem Leaves Extract

1. Collection of Leaves: Fresh Neem leaves were obtained and cleaned from tap water then DI water.
2. Drying of the Leaves: The collected leaves were completely dried to get rid of any moisture.
3. Grinding of Leaves: The completely dried leaves were grinded to form fine powder.
4. Extraction Process: The extracted powder was mixed with DI water and stirred using a magnetic stirrer for 4 hours at 60°C.
5. Filtration Process: After stirring for four hours, the mixture was filtered using Whatman filter paper.



Synthesis of Multi-Metal Oxide Nanoparticles

6. Metal Salt Addition: Certain metal salts were added to the extract. The mixture was kept at 40 °C for one hour using a magnetic stirrer.
7. Metal Precursors: Most of the metal precursors added were Iron Nitrate Nanohydrate (98+% metal basis).

Some others used were:

- Cobalt (II) Nitrate Hexahydrate
- Zinc Nitrate Hexahydrate (Reagent grade 98%)
- Magnesium Nitrate
- Copper (II) Nitrate Trihydrate
- Nickel (II) Nitrate Hexahydrate

8. Observation: Dark brown color appearance was seen which indicates the start of the reduction process.
9. pH Adjustment: Sodium Hydroxide (NaOH) was added drop by drop to get stable pH as 8.
10. Thermal Ripening: The mixture was heated inside the oven at 180 °C for 11 hours.
11. Separation: Centrifugation process was employed to separate the solid phase.
12. Calcination: The resultant material obtained after the separation process is calcined at a particular

stabilized temperature.

Logic for Using Multi-Metal procedure

Logic behind the Use of Multimetal Procedure but Iron is the main element, the introduction of other elements such as Cobalt, Zinc, Magnesium, Copper, and Nickel will bring about some scientific advantages. By mixing these various metal ions, the resultant product will be mixed metal oxide which has improved electron affinity and catalytic efficiency. This mixture will enable doping of the metal oxides and thus introduce imperfections in the crystal lattice resulting in bandgap engineering and improvement in Surface Plasmon Resonance capability. In addition, there will be a blend of characteristics such as high magnetism in Iron and Cobalt with antimicrobial nature in Copper and Zinc.

Characterization via Optical Methods: UV-Vis Spectroscopy

The main technique applied for the purpose of characterization in this study is UV-Vis spectroscopy. Such method implies measurement of the absorbance spectra of light from the synthesized nanoparticles within the wavelength interval of 200 to 800 nm. For this purpose, 10 mg of the prepared nanoparticle powder were dissolved in 100 ml of DI water, and then the high resolution absorbance spectra were acquired.

Experimental Results and Discussion
Two important results were found in the experimental results of the UV-Vis Spectroscopy analysis, which include the absorption peak at 343 nm, and intense absorption of light at 200 nm.

Explanation of the 343 nm Peak (Red Shift)
An absorption peak was found at 343 nm. This absorption peak is known as Surface Plasmon Resonance (SPR) for the multiple metal oxides nanoparticles. The typical absorption peaks of Iron oxide nanoparticles lie between 290-300 nm. However, in the oexperiment, a red shift of 343 nm was found due to the following reasons:

1. Thermal Ripening: Heating the sample for 11 hours at 180 °C produces large particles.
2. Agglomeration: Drying and calcination processes cause the agglomeration of nanoparticles and shift the SPR peak to a long wavelength.

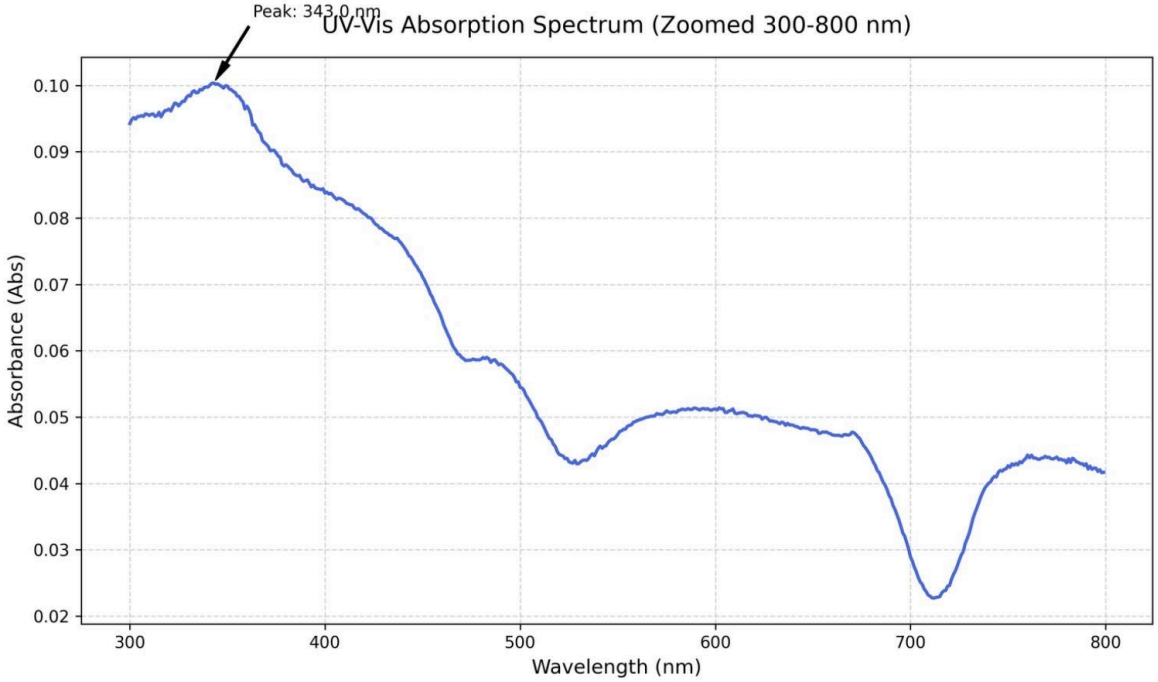


Figure 1: High-resolution UV-Vis Absorption Spectrum (300-800 nm).

Explanation of the 200 nm Peak
Intense absorption of light at 200 nm was observed, and this result can be attributed to the following reasons:

1. Phytochemicals: Neem leaf extract contains organic substances like polyphenol and flavonoid compounds and has absorption properties in deep UV region.
2. Band-Gap Absorption: This phenomenon explains the absorption transition from valence band to the conduction band of the metals.

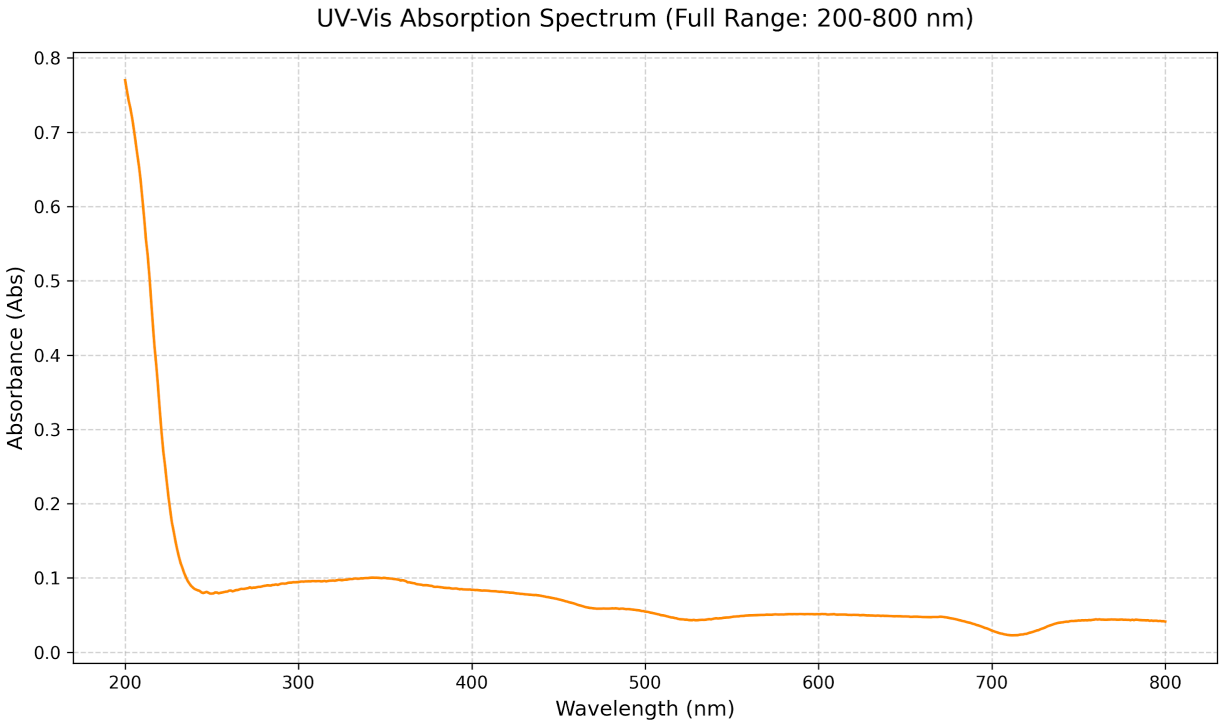


Figure 2 : UV - Visible Absorption Spectrum

Conclusion and Future Scope

Green Synthesis of Multi Metal Oxide Nanoparticles using Neem (Azadirachta indica) Extract has been successfully carried out. Following a well-defined protocol with metal salts such as Iron, Cobalt, Zinc, Magnesium, Copper and Nickel, stable nanoparticle structures have been achieved. Empirical evidence is presented with the help of UV-Visible Spectroscopy.

Scientific facts deduced from analysis of the 10mg/100ml DI water sample show that:
1. The Peak at 343 nm: This proves the presence of nanoparticle with red-shift due to 11 hours thermal ripening at 180°C with particle agglomeration while creating the powder.
2. The Peak at 200 nm: This is the cumulative result of absorption of the phytochemical from Neem and also the fundamental band gap absorption of metal oxide.
Inexpensive and eco-friendly techniques for synthesizing nanomaterials has been found. Further study is required in this area with regards to SEM and XRD analysis. Also, testing for photocatalytic activity in dye degradation and antibacterial application should be carried out.

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